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ALY 6050

Module 4 Project

A Prescriptive Model for
Strategic Decision-making,
An Inventory Management
Decision Model

Introduction

Working as a consultant that is designated the opportunity to work for an organization that sells key engine components, the focus is to find a balance between having ample inventory that meets demand while keeping inventory costs at a minimum. To achieve this balance a decision model must be established that will help identify the best order quantity based off the financial estimates of the inventory. Historical data that shows annual demand of units combined with inventory holding and ordering costs are supplied and will be used in this report that will result in estimating revenue and costs that will lead to maximum profit.

Analysis

Part I

The data provided by the company for evaluation of inventory management indicates that 15,000 units are sold annually and cost \$80 per unit. To hold the units in stock, it costs 18 percent of the cost, equaling \$14.40 each. The other parameter to consider is the ordering cost which after factoring in overhead, clerical work, data processing, and other expenses equals \$220. Minimizing the total inventory cost is the objective and model parameter and the cost per unit and ordering cost are also modeling parameters in this assessment. The decision variables are the holding cost per unit in percentages and values, and the uncontrollable inputs are the demand of the key engine components and the ordering costs.

1. Define Data		
Demand (in units)	15000	Data/Uncontrollable Input
Cost per unit (\$)	\$80	Data/Model Parameter
Holding cost per unit (%)	0.18	Data/Decision Variable
Holding cost per unit in \$	\$14.40	Data/Decision Variable
Ordering cost (\$)	\$220	Data/Uncontrollable Input/Modeling Parameter

The objective of finding the lowest total inventory cost will be based off the costs from the annual holding costs and annual ordering costs. It will also rely on finding the most adequate order quantity in units so ordering costs for the calendar year remain as low as possible. The total costs will be a calculation that adds the model parameters after their individual calculations are completed. These functions are put into a spreadsheet in Excel which can account for making the necessary adjustments when demand and other patterns can change and therefore costs in reordering of units and inventory costs will also be revised (Evans, 2012).

2. Mathematical Functions		
Annual Holding Cost	Demand * Holding Cost Per Unit(\$)	Model Parameter
Annual Ordering Cost	(Demand * Ordering Cost) / Ordering Quantity	Model Parameter
Total Inventory Cost	Annual Holding Cost + Annual Ordering Cost	Objective/Model Parameter
Mathematical Model		
Economic Ordering Quantity (in units)/Reorder Point	677	Decision Variable/Model Parameter
Annual Holding Cost	\$216,000.00	
Annual Ordering Cost	\$4,874	
Total Inventory Cost	\$220,874.42	

The calculations of order quantities and total inventory cost can also be computed and verified using R. The total inventory cost that would be at its lowest for locating maximum profit from units sold from one year and considering holding costs in inventory would be ordering 677 units at a time.

```
> # Calculate EOQ
> EOQ <- calculate_EOQ(D, s, H)
> # Calculate total inventory cost
> total_cost <- calculate_total_cost(D, s, H, EOQ)
> # Print the results
> cat("Economic Order Quantity (EOQ):", EOQ, "\n")
Economic Order Quantity (EOQ): 677.0032
> cat("Total inventory cost:", total_cost, "\n")
Total inventory cost: 9748.846
```

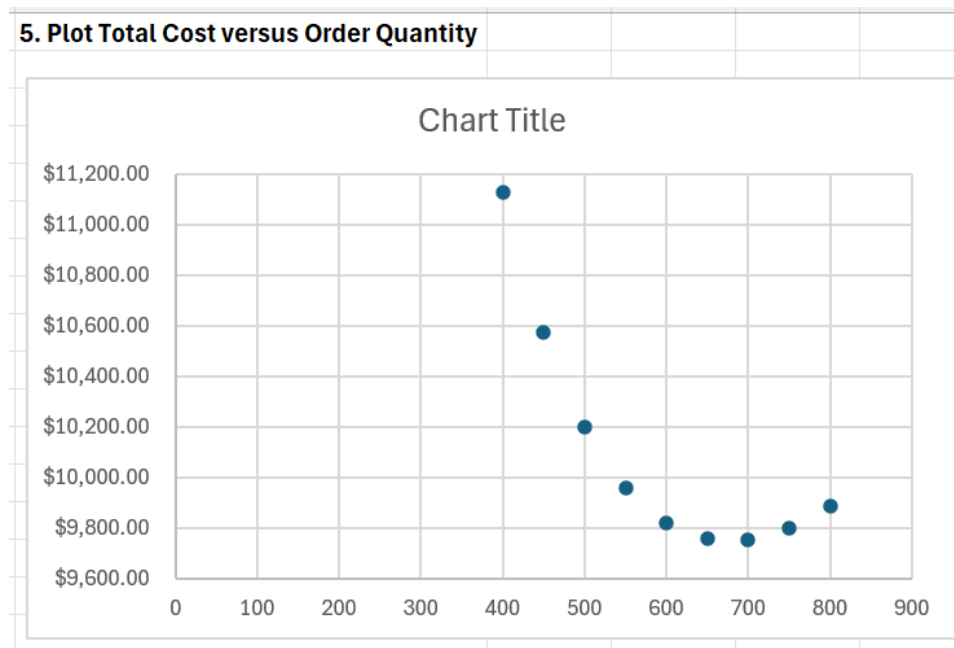
To validate the results of the functions in R, a data table is built to find an approximate order quantity that will formulate the lowest total cost. A range of 400 to 900 with increments of 50 are used with an annual cost calculation provided below (Kaplan, 2020):

Ordering Cost * (Demand / Order Quantity) + Holding Cost Per Unit * (Order Quantity / 2)

The range provided shows the total cost declining as the quantity increases but eventually begins to increase again between 650 and 700 units, indicating that the ultimate order quantity is between those two quantity values.

4. Data Tables	
Order Quantity	Total Cost
400	\$11,130.00
450	\$10,573.33
500	\$10,200.00
550	\$9,960.00
600	\$9,820.00
650	\$9,756.92
700	\$9,754.29
750	\$9,800.00
800	\$9,885.00
850	\$10,002.35
900	\$10,146.67

The data table with the range of order quantities and total costs is shown visually with a scatterplot. An analysis of the data table and plot concludes that there are some benefits to ordering more units per order where total costs can be reduced but only to a certain point. Between 650 and 700 units, the total cost increases, and purchasing additional units loses its beneficial value.



The solver function in Excel is initiated to find the exact order quantity that will lead to the lowest total cost. Considering the range between 650 and 700 units, an estimate of 675 is used in the solver function with the data table range. It is calculated from the inputs provided that the

ultimate order quantity equals 677 units and a minimum total cost of \$9748.85. This validates the past calculations that this is accurate and should be used for future ordering and reordering.

6. Solver	Order Quantity	Minimum Total Cost
	677.0032004	\$9,748.85

A what-if analysis is conducted using the Excel function to measure the vulnerabilities of the total costs if any of the model parameters were revised. As an example, the same order quantity ranges used previously and holding cost rates in a two-way table that displays how the costs were changed if the model parameters were changed. The combination of inputs can look at only one output variable but is effective in showing that the order quantity and holding cost rates both increase the total inventory cost (Evans, 2012). This makes finding the right order quantity so crucial to generating as much profit as possible when knowing the annual demand of the product being sold.

7. What-If Analysis		Order Quantity										
		\$220,874.42	400	450	500	550	600	650	700	750	800	850
Holding Cost Rates	\$9.60	4489.92	5009.35	5526.64	6042.10	6555.99	7068.49	7579.77	8089.94	8599.13	9107.42	
	\$11.20	5182.00	5784.58	6384.86	6983.16	7579.77	8174.87	8768.65	9361.25	9952.77	10543.33	
	\$12.80	5870.47	6555.99	7239.05	7920.00	8599.13	9276.66	9952.77	10627.62	11301.32	11973.98	
	\$14.40	6555.99	7324.27	8089.94	8853.38	9614.88	10374.69	11133.00	11889.95	12645.70	13400.34	
	\$16.00	7239.05	8089.94	8938.08	9783.87	10627.62	11469.58	12309.95	13148.91	13986.59	14823.11	
	\$17.60	7920.00	8853.38	9783.87	10711.89	11637.78	12561.78	13484.13	14404.99	15324.51	16242.81	
	\$19.20	8599.13	9614.88	10627.62	11637.78	12645.70	13651.67	14655.89	15658.57	16659.85	17659.85	
	\$20.80	9276.66	10374.69	11469.58	12561.78	13651.67	14739.51	15825.54	16909.96	17992.92	19074.56	
	\$22.40	9952.77	11133.00	12309.95	13484.13	14655.89	15825.54	16993.32	18159.41	19323.99	20487.20	
	\$24.00	10627.62	11889.95	13148.91	14404.99	15658.57	16909.96	18159.41	19407.12	20653.27	21898.00	

Part II

A simulation of 1000 occurrences is conducted that computes the minimum total cost in R if a triangular distribution of 13000 to 170000 units is implemented. By including the mode of 15000 units, this makes the range a triangular distribution with 13000 as the minimum, 17000 as the maximum, and 15000 as the most likely value in the distribution (Evans, 2012). The expected minimum total cost, the expected order quantity, and the expected annual number of orders are then estimated by using the 95 percent confidence interval.

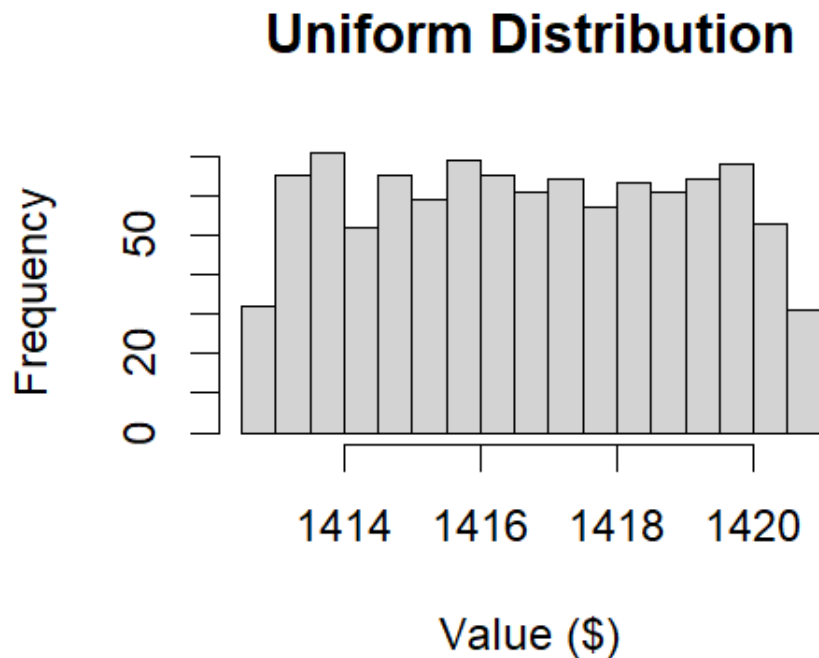
- (i) From the calculations in R the expected minimum total cost is \$1418.18, and the confidence interval is from \$1414.24 and \$1422.12. A uniform distribution is a great fit as a probability distribution because the costs based off the parameters and past data used should remain within a particular range. These results are provided in the histogram below where the bins indicate a consistent trend from the simulation.

```

> #i. Estimate expected minimum total cost
> expected_min_total_cost <- mean(total_costs)
> confidence_interval_min_total_cost <- t.test(total_costs)$conf.int
> # Print results
> cat("Expected Minimum Total Cost:", expected_min_total_cost, "\n")
Expected Minimum Total Cost: 1418.179
> cat("Confidence Interval (95%) for Minimum Total Cost:", confidence_interval_min_total_cost, "\n\n")
Confidence Interval (95%) for Minimum Total Cost: 1414.241 1422.116

> # Probability distribution that best fits its distribution (Uniform Distribution)
> # Set seed for reproducibility
> set.seed(123)
> # Generate 1000 random numbers from a uniform distribution
> uniform_data <- runif(1000, min = 1412.719, max = 1420.749)
> # Plot histogram of the uniform distribution
> hist(uniform_data, breaks = 20, main = "Uniform Distribution", xlab = "Value ($)", ylab = "Frequency")

```



- (ii) From the calculations in R the expected order quantity is 677, and it has a confidence interval that is from 14,928 and 15,026. A normal distribution was used and displayed in a histogram that is shown below. From the simulations, the quantities have a normal bell-shaped curve indicating a normal distribution with the continuous data used.

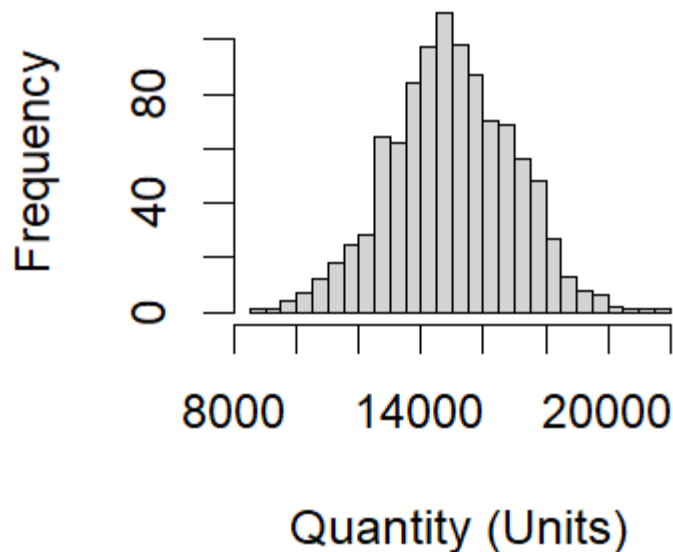
```

> #ii. Estimate expected order quantity
> expected_order_quantity <- sqrt(2 * mean(demand_samples) * ordering_cost / holding_cost_rate)
> confidence_interval_order_quantity <- t.test(demand_samples)$conf.int
> #Print results
> cat("Expected Order Quantity:", expected_order_quantity, "\n")
Expected Order Quantity: 677.4494
> cat("Confidence Interval (95%) for Order Quantity:", confidence_interval_order_quantity, "\n\n")
Confidence Interval (95%) for Order Quantity: 14969.89 15069.67

> # Probability distribution that best fits its distribution (Normal Distribution)
> # Generate random numbers from a normal distribution with a different mean and standard deviation
> custom_random_numbers <- rnorm(1000, mean = 15000, sd = 2000)
> # Plot histogram of the uniform distribution
> hist(custom_random_numbers, breaks = 20, main = "Normal Distribution", xlab = "Quantity (Units)", ylab = "Frequency")

```

Normal Distribution



(iii) From the calculations in R of the expected annual number of orders, the amount is 408, and it has a confidence interval that is from 14,928 and 15,026. An exponential distribution was provided due to the possibilities of different waiting times between orders. To verify these calculations the summary statistics were computed and show the minimum and maximum, the first and third quartiles, the median, and the mean. While it has a few outliers, most occurrences from the simulations are close to the mean.

```
> #iii. Estimate expected annual number of orders
> expected_annual_orders <- sqrt(2 * mean(demand_samples) / holding_cost_rate)
> confidence_interval_annual_orders <- t.test(demand_samples)$conf.int
> # Print results
> cat("Expected Annual Number of Orders:", expected_annual_orders, "\n")
Expected Annual Number of Orders: 407.9384
> cat("Confidence Interval (95%) for Annual Number of Orders:", confidence_interval_annual_orders, "\n")
Confidence Interval (95%) for Annual Number of Orders: 14928.02 15026.46
> # Probability distribution that best fits its distribution (Exponential Distribution)
> # Set seed for reproducibility
> set.seed(456)
> # Calculate the mean of the triangular distribution
> mean_triangular <- expected_annual_orders
> # Calculate the rate parameter for the exponential distribution
> lambda <- 1 / mean_triangular # Lambda is the reciprocal of the mean
> # Generate random values from the exponential distribution
> simulated_values_exponential <- rexp(1000, rate = lambda)
> # Print the summary statistics
> summary(simulated_values_exponential)
   Min.    1st Qu.    Median      Mean   3rd Qu.      Max.
 0.4288  116.3273  276.0373  389.3943  528.1514 2770.6334
```

Conclusion

The first part of the project conducts several analysis tests that demonstrate how the company can generate enough profit while keeping costs down. These values were tested and verified and give the CEO of the company the opportunity to see how over one year these values can create revenue and adjust to changing demands and supply concerns. An example of this was in the second part of the project where demand was lower and higher than expected and how the company can adapt its model by looking at different probability distributions through simulation testing. These tools at their disposal will have them equipped to handle any scenario in a market that is always changing.

References

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Evans, J. (2012). Statistics, Data Analysis, and Decision Modeling, 5th Edition. Pearson Publishing.

Kaplan Financial Knowledge Bank. (2020). *Chapter 6: Order quantities and reorder levels*. Kaplan Financial Limited. <https://kfknowledgebank.kaplan.co.uk/acca/chapter-6-order-quantities-and-reorder-levelsv2>